IoT-Enabled Hydroponics

AMAL C THOMAS

June 26, 2024

The Rise of Smart Agriculture

- Growing demand for sustainable and efficient farming practices.
- IoT (Internet of Things) revolutionizing agriculture with data-driven insights.

Hydroponics: Growing Smarter

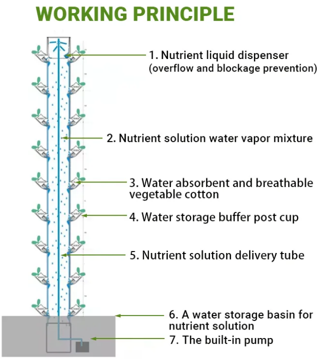
- Water-efficient method for growing plants without soil.
- Offers precise control over nutrients and environment.

Introducing Project

- An IoT-enabled hydroponics system for optimized plant growth.
- Utilizes sensors, automation, and data visualization for smarter farming.

Traditional Hydroponics Systems

- Relies on a basic pump system for nutrient delivery.
- Constant pumping can lead to overwatering and nutrient runoff.
- Limited monitoring capabilities hinder adjustments for optimal plant health.



Smart Hydroponics System

- This project builds upon traditional systems for improved efficiency and plant health.
- Features intelligent irrigation based on sensor data.
- Integrates environmental monitoring for a holistic view of plant needs.

Intelligent Irrigation

Watering When Your Plants Need It Most

- Traditional systems: Constant pumping can lead to water waste.
- Our solution: Intelligent irrigation based on real-time data.
- Scheduled pumping every 30 minutes for 3 minutes.
- Humidity sensor triggers additional pumping only if needed (below 90% humidity).
- **Benefits:**
  - Reduced water waste.
  - Optimized nutrient delivery based on plant needs.

Importance of pH and EC in Hydroponics

- **pH (Potential of Hydrogen):**
  - Determines the acidity or alkalinity of the nutrient solution.
  - Essential for nutrient availability and uptake by plants.
  - Ideal pH range varies with plant type, generally between 5.5 and 6.5 for most hydroponic crops.
  - Incorrect pH levels can lead to nutrient deficiencies or toxicities, stunting plant growth.
- **EC (Electrical Conductivity):**
  - Measures the concentration of dissolved salts (nutrients) in the nutrient solution.
  - Indicates the overall nutrient strength available to plants.
  - Ideal EC levels vary with plant type and growth stage.
  - High EC can cause nutrient burn, while low EC can lead to nutrient deficiencies.

Plant EC and pH Values

Crops	EC (mS/cm)
Cucumber	1.7-2.2
Lettuce	0.8-1.2
Onions	1.4-1.8
Potato	2.0-2.5
Spinach	1.8-2.3
Tomato	2.0-5.0
Carrots	1.6-2.0

EC Values of Various Plants

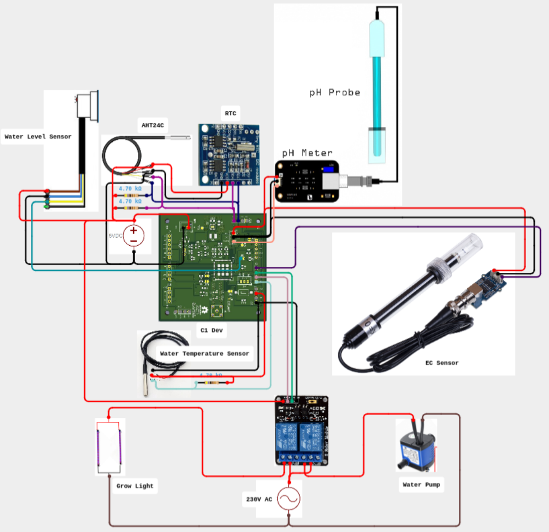
Crops	pH
Cucumber	5.8-6.0
Lettuce	5.5-6.5
Onions	6.0-7.0
Potato	5.0-6.0
Spinach	5.5-6.6
Tomato	5.5-6.5
Carrots	5.8-6.4

pH Values of Various Plants

Integrated Sensors for Real-Time Data

- **pH Sensor:** Measures acidity/alkalinity of nutrient solution, vital for nutrient uptake.
- **EC Sensor:** Measures electrical conductivity (EC), an indirect indicator of nutrient concentration.
- **DS18B20 Sensor:** Measures the temperature of the nutrient solution, influencing EC.
- **AHT2415C Sensor:** Monitors humidity, impacting plant transpiration and nutrient needs.
- **Water Level Sensor:** Monitors the level of nutrient solution in the reservoir, preventing pump operation when empty.

Circuit Diagram



Data Transmission and Visualization

- Real-time sensor data is crucial, but we need a way to utilize it effectively.
- Our system transmits data wirelessly using the LoRaWAN protocol.
- LoRaWAN offers long-range, low-power communication for sensor networks.
- Data visualization with Grafana transforms raw data into actionable insights.
- Grafana creates interactive dashboards displaying sensor readings (pH, EC, humidity, temperature) over time.

Benefits:

- Easy monitoring of various environmental factors from a remote location.
- Identify trends and patterns to optimize plant growth conditions.
- Make informed decisions based on real-time data visualization.

Benefits of the Smart Hydroponics System

- **Increased Efficiency:**
  - Reduced water waste through intelligent irrigation.
  - Optimized nutrient delivery based on real-time sensor data.
- **Improved Plant Growth Potential:**
  - Precise control over environmental factors for ideal growth conditions.
  - Early detection of potential issues allows for proactive adjustments.
- **Enhanced Monitoring and Control:**
  - Real-time data visualization for remote monitoring of plant health.
  - Ability to make informed decisions based on sensor readings.
- **Reduced Labor Requirements:**
  - Automated irrigation and data collection minimize manual tasks.

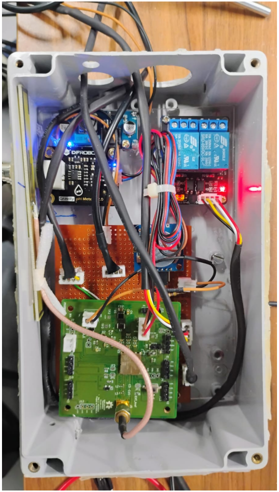
Applications of the Smart Hydroponics System

- **Urban Farming:** Enables efficient and productive plant cultivation in urban environments with limited space.
- **Personal Home Gardens:** Provides individuals with a convenient and controlled way to grow fresh, healthy produce at home.
- **Educational Purposes:** Serves as a valuable tool for educational institutions to teach students about hydroponics and smart agriculture.
- **Research and Development:** Creates a platform for researchers to explore the optimization of plant growth conditions and develop new farming techniques.

Deployment: Planted Spinach

- **Spinach Planting:**
  - Carefully transplanted spinach plants into the hydroponics system.
  - Ensured optimal spacing for each plant to promote healthy growth.
- **Initial Setup:**
  - Monitored environmental conditions including pH and EC levels.
  - Adjusted nutrient solution to match spinach requirements.
  - Configured sensors for continuous monitoring and data collection.

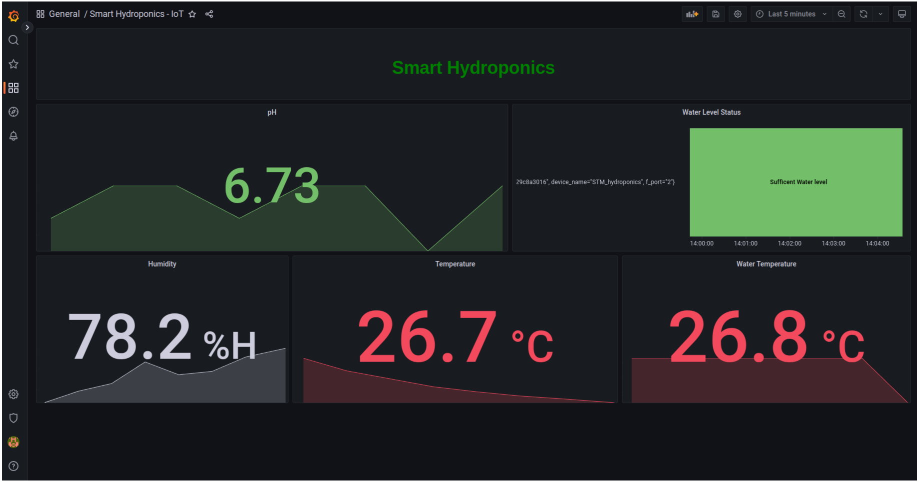
Deployment



Deployment: Planted Spinach

- **Growth Monitoring:**
  - Regularly checked growth progress and adjusted settings as needed.
  - Used data from sensors to optimize water and nutrient delivery.
  - Ensured consistent lighting and temperature for healthy growth.
- **Data Visualization:**
  - Real-time data visualized on Grafana dashboard.
  - Grafana dashboards display sensor readings such as pH, EC, humidity, and temperature.
  - Utilized data insights to make informed decisions for plant care.
- **Expected Outcomes:**
  - Healthy and robust spinach plants with optimal nutrient uptake.
  - Improved yield and quality of spinach due to precise control.

Data Visualization



Visualization

Thank You